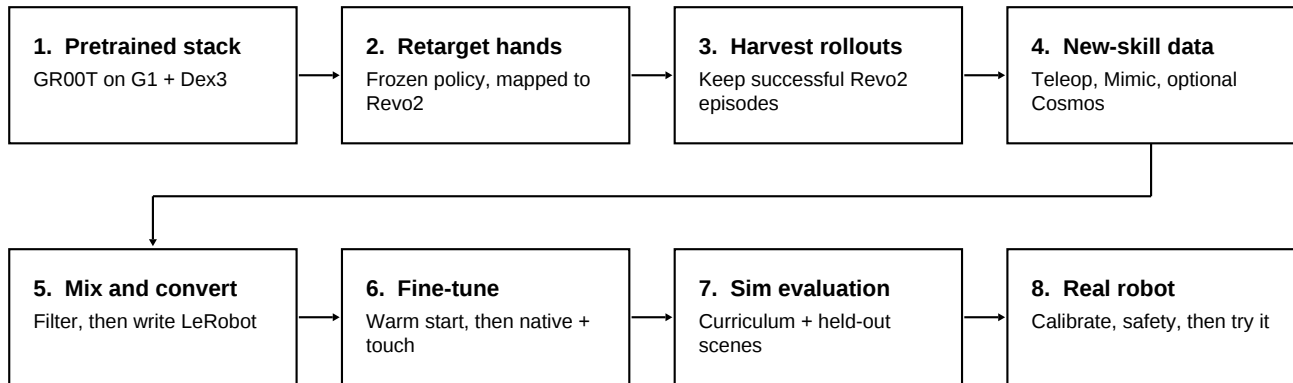


Take the GR00T / standing-WBC stack we already have on the Unitree G1 and teach a new skill on BrainCo Revo2 Touch hands: pick the object with the right hand, pass it to the left, put it down. The giving hand should open when the other hand feels the object, not on a timer. What follows is the sequence, start to finish. No scores, just the process.

Figure 1. The pipeline

Left to right along each row, then down from box 4 into box 5.



The task itself

Right hand reaches and grasps. Both wrists come together in front of the chest. Left hand closes when it has contact. Right hand opens only after that catch is felt. Left hand places the object. The object has to stay in a hand the whole time — we do not teleport it from one palm to the other.

What we keep and what we change

Keep the vision backbone, the language grounding, the arm motion the checkpoint already has, and the standing balance controller. Change the hands and the action around the exchange. Harvested rollouts give us Revo2 pick-and-place. Teleop and Mimic give us the handover. Real robot data is only for whatever still looks wrong after calibration.

Each step, briefly

1. Pretrained stack

Start from the public GR00T checkpoint on the G1 with Dex3 hands and the standing whole-body controller. This step is only to check that vision, language, the arms and balance already work on pick-and-place. If that loop is broken, there is no point changing the hands or the task.

2. Retarget the hands

Swap Dex3 for BrainCo Revo2 Touch and leave the policy frozen. A mapping layer turns source-hand commands into Revo2 motor commands, and turns Revo2 joint state back into the format the checkpoint expects. Same model, new hands.

3. Harvest those rollouts

Run the frozen, retargeted policy in a few parallel sim environments and save the episodes that finish pick-and-place, plus the near-misses. Those clips already show Revo2 fingers in the camera and already use Revo2 contact. They cannot teach the handover, but they are cheap embodiment data.

4. Collect the new skill

Handover is not pick-and-place, so we still need new demonstrations. Record a small set of teleoperated handovers in sim, grow them with Isaac Lab Mimic (left hand only closes after the right has arrived; right only opens after the left has a grip), and optionally re-render a few with Cosmos for lighting.

5. Mix and convert

Throw out incomplete or unsafe episodes. Convert what remains to LeRobot, including Revo2 joints and fingertip Touch. Put harvested pick-and-place, Mimic handovers and later a little real teleop in the same pool. Each source covers something the others do not.

6. Fine-tune, twice

Keep the vision-language backbone frozen. First pass: stay on the Dex3 action interface and the retargeter, and only train the projectors and the action head. Second pass: a native Revo2 head plus Touch, so the policy can open the giving hand when it feels the catch, not when the head camera guesses (the hands hide the object).

7. Evaluate in simulation

Do not jump straight to the full task. Frozen baseline first, then grasp on Revo2, then the full handover, then the tactile version, then a mix that includes real clips. Look at each part of the exchange on scenes the training set did not use. Also run it without Touch, and without the harvested pick-and-place, so we know what each piece is for.

8. Put it on the real robot

Calibrate the camera, the hand map and Touch. Use the same observations and actions as in sim. Keep a safety layer that can stop the robot even if the policy is wrong. Try hands on the bench, then arms, then a grasp, then the full handover. Record real teleop only for whatever calibration did not fix.